

Intelligent Cane

Neural-Nexus / Track A / Technical progress / 5 October 2026

Dulnith, Thenul, Chamith, Suneth

PROBLEM

White-cane users may benefit from supplementary feedback about obstacles within a sensor's field of view. This prototype explores affordable tactile and audible feedback with local processing.

PROPOSED SOLUTION

An ESP32-C3 reads an ultrasonic sensor and increases vibration and beep urgency below 60 cm. An inertial sensor provides a cane orientation/impact alarm. This is a bench proof of concept, not validated human-fall detection.

ARCHITECTURE



USB / BLE NUS telemetry to a terminal. Planned: phone location and caregiver delivery.

CURRENT PROGRESS

The team connected the board, ultrasonic sensor, IMU, motor and buzzer. Active firmware implements feedback, the cane alarm and BLE telemetry. Physical operation still needs recorded bench evidence.

TECHNOLOGY STACK

ESP32-C3 SuperMini, HC-SR04, GY-521/MPU6050, motor module and buzzer. Embedded C++, Arduino-ESP32 3.3.2, Adafruit MPU6050, I2C, LEDC PWM and BLE GATT/NUS. Listed base electronics: **LKR 2,800**, excluding unpriced divider, power, cane and enclosure.

TESTING

ESP32-C3 compilation passed on 5 October. Existing host tests passed for boundaries, invalid inputs, monotonic feedback, polarity, beep transitions and timer rollover. Physical accuracy, latency, BLE range and battery runtime remain unmeasured in the evidence reviewed.

LIMITATIONS

One sensing direction; missing Echo is ambiguous. Cane movement can trigger the alarm. No downward sensor, caregiver delivery or AI implementation. Verify Echo level conversion and actuator drivers physically. No guaranteed response-time or battery-runtime claim.

NEXT STEP

Record repeated distance and tilt tests, secure wiring/mounting and improve fault indication. Then complete one phone-to-caregiver alert path. Camera/AI remains an optional later extension.

Repository: github.com/Neural-Nexus-IoTrix-2-0/intelligent-cane

Evidence: [docs/test-logs/2026-10-05-software-validation.md](#)